BRNO UNIVERSITY OF TECHNOLOGY FACULTY OF ELECTRICAL ENGINEERING AND COMMUNICATION

Robotics and AI

MPC-MAP Assignment No. 5 - Report

Author: Tomáš Špelda

Date: 24. 04. 2026

Task 1

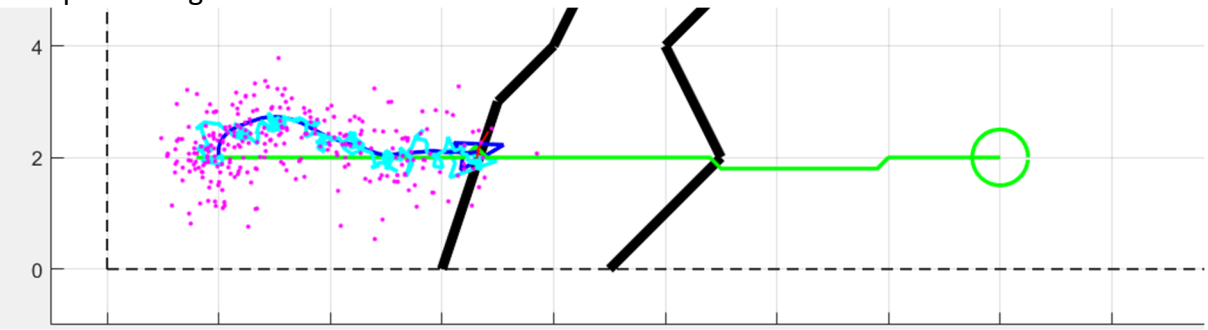
Algorithm

- A* algorithm was write
- World coordinates are converted to grid index
- Start and goal are mapped to grid cells

Core

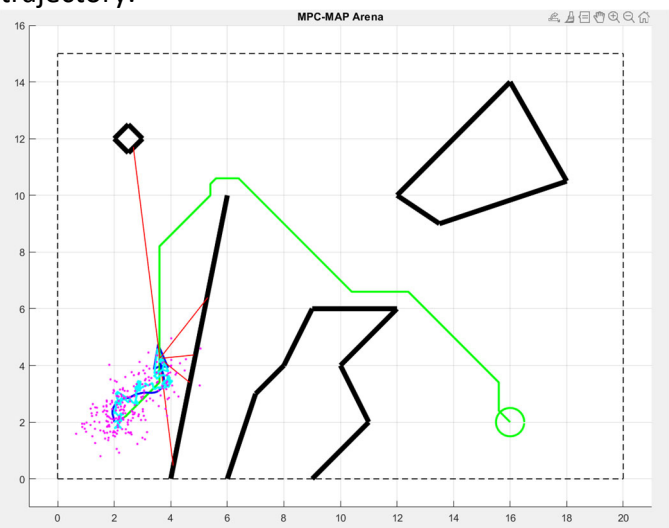
- Uses a priority queue to explore lowest-cost nodes first
- Uses Euclidean distance
- Explores 8 directions (cost 1 straight, sqrt(2) diagonal)
- Reconstructs the path and converts it back to world coordinates

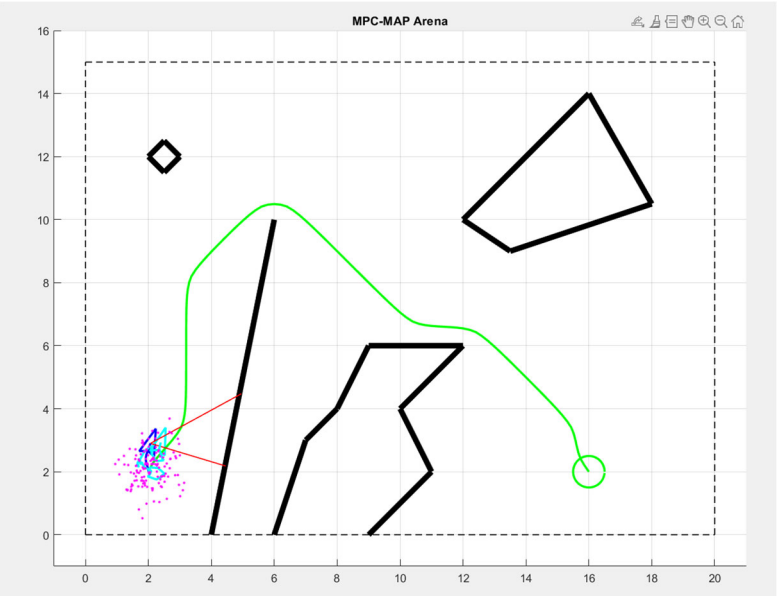
Find path but ignores obstacles.



Task 2

An iterative smoothing method was used to refine the raw A* path and improve the trajectory.



Task 3

A smoothing process was applied iteratively to the A* path to make the trajectory smoother and cleaner.